



Predicting Heart Disease

A Statistical Classification Study for Clinical Risk Screening

AAI-500 · Applied Probability and Statistics for AI

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github.com/pdhinaka/AAI500-Final-Project



What We'll Cover

1	Introduction & Business Problem	Pranav
2	The Dataset	Mina
3	Data Cleaning & Preparation	Mina
4	What the Data Showed (EDA)	Mina
5	Building the Model	Pranav
6	How Well It Works	Sristi
7	Limitations & Considerations	Sristi
8	Conclusion & Recommendations	Pranav

1



Introduction & Business Problem

OWNER Pranav

What if the data a hospital already collects could help catch heart disease earlier?



THE PROBLEM

Heart disease is the world's leading cause of death. Diagnosing it means piecing together many tests, the process is slow, and some at-risk patients are missed.



OUR SOLUTION

A screening aid that reads 11 routine measurements a hospital already collects, flags the patients most likely to have heart disease and shows the reasons why.



THE PAYOFF

Earlier flags mean earlier treatment, fewer missed cases, and smarter use of doctors' time, with no new tests or equipment required.



The Dataset

OWNER Mina

303

PATIENTS (RAW)

Records after cleaning: **297 patients**

14

ATTRIBUTES

2

OUTCOMES (DISEASE / NO DISEASE)

Source

UCI Heart Disease (Cleveland). Each row is one patient.
The target was simplified to a yes / no diagnosis.

PLAIN-LANGUAGE DESCRIPTION

- Patients are primarily **middle-aged and older adults**.
- Cholesterol and blood pressure exhibit substantial variability.
- The very large cholesterol range (126–564) suggests potential outliers.
- Continuous variables span different scales, indicating scaling will likely be required before modeling.

3



Data Cleaning & Preparation

OWNER Mina

Handled missing data

6 of 303 patients were missing values. We removed them, leaving 297 complete records (about 2%).

Defined the outcome

The original 0-4 severity scale was collapsed to a simple disease / no-disease label.

Prepared for modeling

Categories were converted to numeric form and measurements put on a common scale.

YOUR TALKING POINT

The dataset required minimal cleaning, retained nearly all patient records, and was prepared using standard machine-learning practices to ensure reliable model training and evaluation.



Strongest Signals of Heart Disease



Lower Maximum Heart Rate (thalach)

- Patients with heart disease reached lower peak heart rates during exercise.
- Suggests reduced cardiovascular performance.



Higher ST Depression (oldpeak)

- Heart disease patients showed more abnormal ECG changes during exercise.
- Indicates possible reduced blood flow to the heart.



Chest Pain Type (cp)

- Certain chest pain patterns were much more common in patients with heart disease.
- One of the strongest indicators of cardiac problems.



Major Vessels (ca) & Thalassemia Test (thal)

- More blocked vessels and abnormal stress-test results were strongly associated with disease.
- These features provided the clearest separation between healthy and diseased patients.

KEY FINDING #1 Maximum Heart Rate (thalach) & ST Depression (oldpeak)

Variable	p-value
thalach	6.1×10^{-14}
oldpeak	3.4×10^{-13}

Variables with p-values below 0.05 are statistically associated with heart disease status.

KEY FINDING # 2 Chest pain type (cp), exercise-induced angina (exang), number of major vessels (ca), and thalassemia category (thal) typically demonstrate the strongest associations with heart disease.

Variable	Chi-Square	p-value
thal	82.46	1.24×10^{-18}
cp	77.28	1.18×10^{-16}
ca	72.30	1.37×10^{-15}
exang	50.94	9.51×10^{-13}



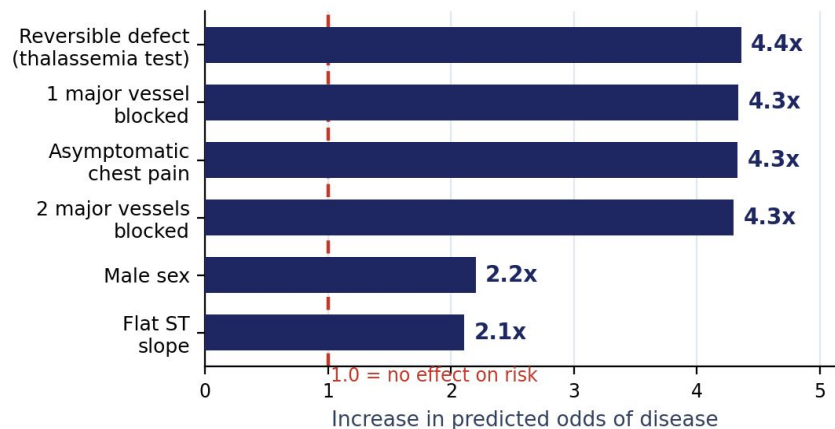
Why logistic regression

It gives a clear yes / no prediction and shows which factors push risk up or down. That interpretability is exactly what a clinical setting needs.

How we set it up

Trained on 80% of patients (237) and tested on the other 20% (60) the model had never seen. Measurements were put on a common scale first.

Factors that most increase predicted risk



11 clinical factors · 80 / 20 train-test split · around 83% test accuracy

6



How Well It Works

OWNER Sristi

[83.3%]

ACCURACY

[75.0%]

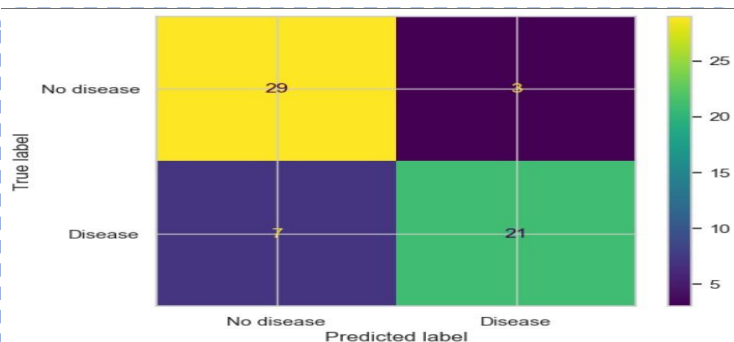
SENSITIVITY

[90.6%]

SPECIFICITY

[0.929]

AUC



Confusion matrix

WHAT THE NUMBERS MEAN

Accuracy (83.3%) : The model correctly classified 83.3% of patients.

Sensitivity(75%) : The model identified 75% of patients who actually had heart disease.

Specificity(90.6%) : The model correctly classified 90.6% of healthy patients.

AUC (0.929) : The model shows excellent ability to distinguish between disease and non-disease patients.



Limitations & Considerations

OWNER Sristi

SMALL DATASET

The dataset only contain 297 patients after cleaning. A sample size may not capture all patient variations and can limit how well the model performs on new data.

COST OF MISSED CASES

The model missed 7 patients with heart disease which is false negative. Missing true disease can delay diagnosis and treatment, making this the most serious type of error for a screening tool.

GENERALIZABILITY

This model was trained using single heart disease dataset and may not represent all patients populations. It would require additional testing on larger and more diverse datasets before clinical use.



Conclusion & Recommendations

OWNER Pranav

WHAT WE FOUND

The model predicts heart disease with around 83% accuracy. Its strongest risk factors match known clinical indicators, so it is capturing real signal, not noise.

WHAT WE RECOMMEND

Use a tool like this as a screening aid that flags higher-risk patients for follow-up. It supports a clinician's judgment, it does not replace it.

WITH MORE TIME / DATA

Collect a larger and more diverse patient set, tune the model to catch more true cases, and compare it against other models.



Team Contributions

Required slide — each member's name and their contributions to the project.



Mina Habib

DATA & EDA

*summary stats, distribution plots,
correlation analysis, categorical and
continuous predictor comparisons
against the target*



Pranav Dhinakar

MODELING & REPORT

*Modeling: feature selection, encoding,
split, scaling, logistic regression,
coefficients*



Sristi Ghimire

EVALUATION

*Model Evaluation, confusion matrix,
AUC analysis, sensitivity and specificity,
and limitations*